1. Momentum

Momentum is mass times velocity and is a vector in the velocity's direction. At the same speed, greater mass gives greater momentum magnitude. Choose a positive direction before comparing collisions and assign negative signs to motion in the opposite direction. Unlike kinetic energy, momentum depends on velocity to the first power rather than speed squared.

$$\vec{p} = m\vec{v}$$

p: momentum (kg m s^{-1}); m: mass (kg); v: velocity (m s^{-1}). Momentum is a vector; non-relativistic motion.

2. Conservation of momentum

A system's total momentum is unchanged during an interaction if the net external impulse is zero. Add the signed momenta of all included objects using one direction convention. Objects sticking together share a final velocity. Momentum conservation applies to both elastic and inelastic collisions and does not imply conservation of kinetic energy.

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

No resultant external impulse during the interaction. m_A, m_B: masses; u: initial velocity; v: final velocity. Signed components along one axis; applies to elastic and inelastic collisions.

3. Elastic and inelastic collisions

An elastic collision preserves total kinetic energy. In an inelastic collision, some kinetic energy becomes deformation, internal energy or sound. In a completely inelastic collision the objects stick together. Both conserve momentum when external impulse is negligible.

4. Resultant force and rate of change of momentum

Resultant force is the rate of change of momentum. Over an interval, average resultant force equals final minus initial momentum divided by elapsed time. Use signed velocities when an object rebounds. For the same momentum change, increasing the interaction time reduces the average force, explaining how cushioning reduces impact force.

$$F_{\text{net, av}} = \frac{\Delta p}{\Delta t} = \frac{mv - mu}{\Delta t}$$

Average resultant force (N); Δp: momentum change (kg m s^{-1}); Δt: s. Last expression requires constant mass. Instantaneous force is the instantaneous rate of change of momentum.

5. Impulse and force-time graphs

The impulse of the resultant force equals the change in momentum and the signed area under the resultant-force-time graph. For a varying force, find impulse from the graph and divide by duration to obtain average force. Positive and negative areas represent opposite directions; N s is equivalent to kg m s^{-1}.

$$\Delta p = \int_{t_1}^{t_2} F_{\text{net}}\,\mathrm{d}t$$

Impulse equals the signed area under the resultant force–time graph (N s = kg m s^{-1}). Integral notation represents this area rule.